SOP-CONFIGURATION MANAGEMENT

[CONFIDENTIAL]
[UTTHUNGA]
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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Prepared By & Date	Reviewed By & Date	Approved By & Date
01	0.1	Draft	Implementation of the Integrated Management System	Sushmitha S 04-Jul-22		
02	Issue B Version 1.0	Revised the Document to suit the requirements for implementation of the integrated management system			Megha K 06-Jul-22	Rama G 08-Jul-22
03	Issue B Version 1.1	Section 9.4.2 & 9.4.3	Criteria for raising CR added	Anitha N 8-Nov-23	Santosh V 9-Nov-23	Santosh V 9-Nov-23
04	Issue B Version 1.2	9.2.1, 9.3 and 9.4	Baseline procedure, CM folder structure, Access to configuration area	Anitha N 06-Dec-23	Santosh V 10-Dec-23	Santosh V 10-Dec-23
05	Issue B Version 1.3		Separation of Configuration Management & Change Management	Sachin Aramani 05-Jan-24	Anitha N 9-Feb-24	Santosh V 9-Feb-24
06	Issue B Version 1.4	Document formatting, font style to Poppins and alignment	Updated as per document guidelines	Aniket Parab 20-Dec-2024	Anitha N 23-Dec-2024	Santosh V 31-Dec-2024
7	Issue C Version 1.0	Review of document	New version update of IMS	Anitha N 01-Aug-25	Santosh V 08-Aug-25	Sanmangala M S 14-Aug-25

1. Purpose

To define a procedure to identify, control and trace the Configuration Items during all the phases of the project development and support functions.

2. Scope

This applies to all the configuration items identified in the project / support group activities.

3. Abbreviations

a. PM: Project Manager

b. CM: Configuration Manager

c. CI: Configurable Item

d. CSI: Customer Supplied Item

e. CRF: Change Request Form

f. CCB: Change Control Board

g. CRR: Change Request Report

h. PP: Project Plan

4. Terms and Definitions

a. NIL.

5. Responsibilities

a. Project Manager shall maintain documents and records related to Integrated Management

b. Configuration Manager shall be responsible for identifying CI & CSI

c. Project Manager shall establish Baseline

d. Project Manager is responsible for maintaining Folder structure.

e. Developer is responsible for their respective CI and its versions

f. Testers are responsible for their respective CI and its versions

6. RACI

Process	Responsible	Accountable	Consulted	Informed
Configuration Management	Project team members	Project Head	Quality Team	Customer / Top Management

Document / Record Maintenance	Project Manager (PM)	Configuration Manager (CM)	Quality Team	Tech Lead, Developers, Testers
Identification of CI	Configuration Manager (CM)	Project Manager (PM)	Quality Team	Tech Lead
Baseline Establishment	Project Manager (PM)	Configuration Manager (CM)	Quality Team	Tech Lead, Developers, Testers
Configuration Audit	Configuration Manager (CM)	Project Manager (PM)	Quality Team	Tech Lead
Folder Structure Maintenance	Project Manager (PM)	Configuration Manager (CM)	Quality Team	Tech Lead, Developers, Testers
Access Control	Configuration Manager (CM)	Project Manager (PM)	Quality Team	Tech Lead, Developers, Testers
Version Control Implementation	Project Manager (PM)	Configuration Manager (CM)	Quality Team	Tech Lead, Developers, Testers

7. SIPOC

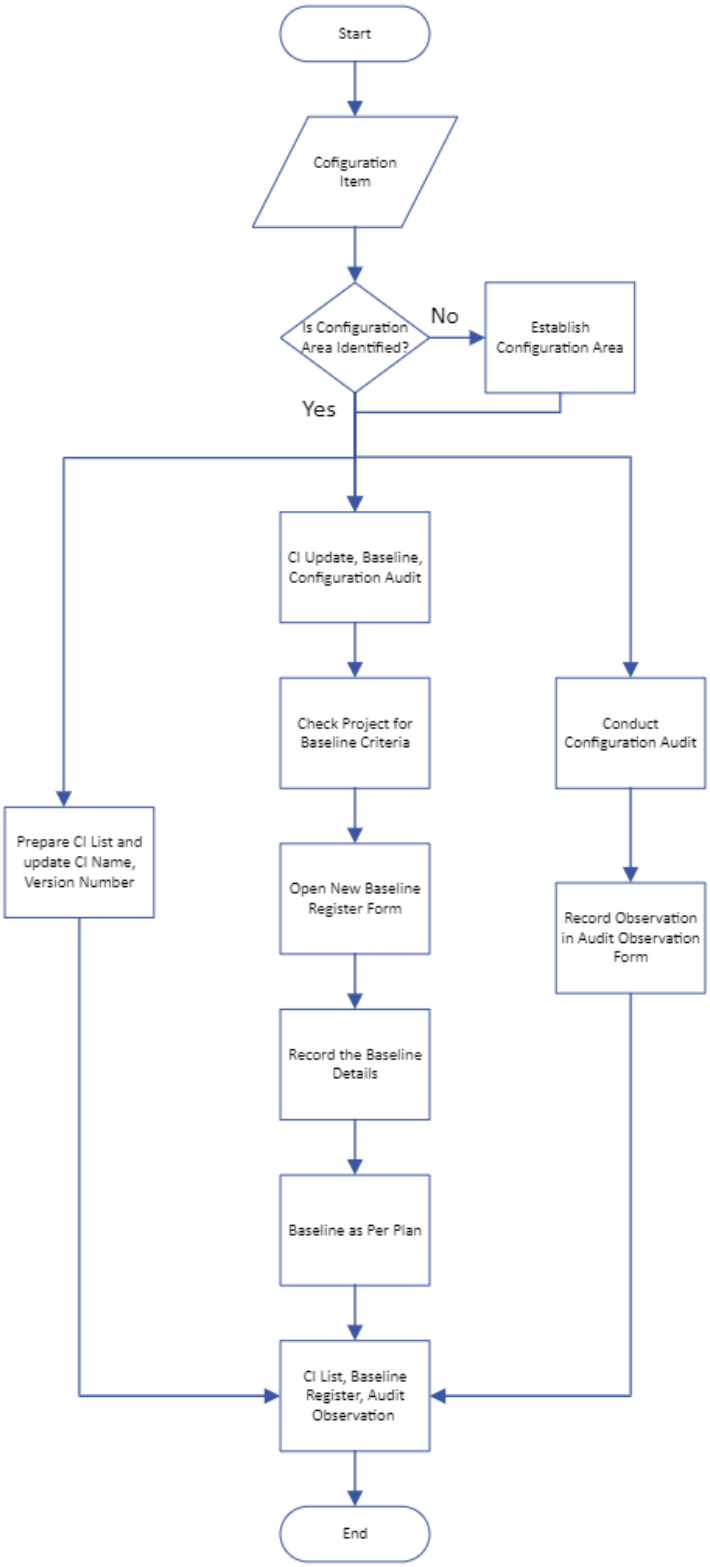
Supplier	Input	Process	Output	Customer
Project Manager	Project artifacts	Document / Record Maintenance	Documents, Quality Records, Project documents, Risk Register, Customer Supplied items, Tools, Source code, Executables, Specified Hardware /	Customer, Project Team, Quality Team

			Software, Development and Target Environments, Training Materials, Support Group Documents	
Project Team	Documents, Quality Records, Project documents, Risk Register, Customer Supplied items, Tools, Source code, Executables, Specified Hardware / Software, Development and Target Environments, Training Materials, Support Group Documents	Identification of CI	Identified CI for Baseline	Customer, Project Team, Quality Team
Project Team	Identified CI for Baseline	Baseline Establishment	Baseline and documented baseline register	Customer, Project Team, Quality Team
Quality Team	Baseline register, Identified CI, CM Checklist	Configuration Audit	Audit Observation Form	Project Team, Quality Team
Quality Team	Recommended Folder structure	Folder Structure Maintenance	Maintained Folder structure	Customer, Project Team, Quality Team

IT Team, Customer	Team member list	Access Control	Role wise access control for repository	Customer, Project Team, Quality Team
Quality Team, IT Team, Customer	Identified CI for Baseline	Version Control Implementation	Implemented version control	Customer, Project Team, Quality Team

8. Overall Process Flow Chart

a) Configuration Management



9. Process Description

9.1 Configuration Management

a. Document / Record Maintenance

- The Project Manager (PM) or person designated by PM shall maintain the quality records related to configuration management in the project and similarly support function head shall maintain respective support function records as soft copy in the configuration area.
- Configuration Report generated from Version control tool.

b. Identification of Configuration Item (CI)

- The Configuration Management Plan (CMP) in the Project Plan / Support Group Plan shall clearly define the CI, which will be a part of the project.
- The CI may be of the following types: -
  - Project documents
  - Risk Register
  - Customer Supplied items.
  - Tools
  - Source code
  - Executables
  - Specified Hardware / Software
  - Development and Target Environments
  - Training Materials
  - Support Group Documents

9.1.1 Baseline of CI

a. Initial baseline shall be established for all the projects.

b. Baseline may be established during the following phases of the project:

- Requirement Baseline – Established when the requirements specification is completed and signed off.
- Design Baseline - Established when the functional requirements are completely translated to the design specifications.
- Product Baseline:
  - Initial product baseline - established after the programs are coded, reviewed, unit tested and accepted.
  - Release baseline - established after the programs are integrated / system tested and accepted.

- Acceptance baseline - established when the final product is accepted by the Customer.
- Archival baseline - established just before the archival of the project.

c. Project Manager shall ensure that the baseline is done as per PP.

d. The details of Baseline items and changes to it shall be recorded in the Baseline Register.

e. Support group baselining will be done annually

f. Baseline Procedure

- Project repository (GIT/SVN) shall be created to store the project data.
- Support groups to baseline all the respective data annually
- Trunk directory shall be created in the project repository and main development line of code database shall be stored.
- Branch shall be created to store the copy of data/code which everyone will modify.
- All changes shall be made in the branch, and it shall be committed to branch directory
- Changes made in the branch directory shall be merged with trunk.
- Create a copy of the baseline version and label it as the baseline version. This copy should be stored in a secure location and should not be modified. The access for the baseline should be provided only for designated persons.
- After baselining the artifacts/code, all-new changes to be identified and documented since the baseline version was established.
- Evaluate the changes to determine if changes are significant and require approval.
- If changes are significant, changes should be approved by the appropriate stakeholders.
- The approved changes to be updated in the baseline version to reflect the changes.
- Communicate the changes to all stakeholders and ensure that they have access to the updated artifacts/code.

9.1.2 Configuration Management folder structure

Recommended folder structure for the configuration area

<<Project Name>>

<< Source >> [source code, inner folder structure could be as per your project needs]

<< CSI >> [Includes all customer supplied items, software, documents..]

<< Documents >>

<< Project Management >>

Project plan (doc)

Project tracking and status reports (doc) (MOMs)

Project execution document

<< Requirements >>

Requirement Document / Product backlog (doc)

Requirement Traceability Metrics

<< Design >>

Design document

<< Coding >>

Unit test cases document / report

<< Testing >>

Test Plan (doc)

TSR, TRR (docs)

<< Reviews >>

PP review form

Requirement review form

Design review form

Code review form

<< Release >>

Internal Release

External Release

<< Mails >>

approval emails

change request emails

appreciation emails

customer complainta/ concerns emails

<< Quality Management >>

<< AuditsAndFindings >> (<<IQA / CMA observations and NCs)

<< Metrics >>

Scrum metrics

Defect metrics

Effort variance

Schedule variance

9.1.3 Configuration Audit

a. Project Manager shall ensure that configuration audit is scheduled in the project plan as part of Configuration Management Plan.

b. Audit for Support functions shall be combined with Internal Audits

c. The audit shall be conducted, and observations shall be recorded in the Audit Observation Form.

d. This activity ensures that:

- All change requests are recorded, analyzed and appropriate action is taken.
- Details of the affected configurable items, current/ new versions and test status have been updated in change request form.
- The CI which has been reviewed and approved is versioned and checked in with predefined version guideline.

e. Ensure the customer supplied items are returned at the time of project closure.

f. Baselines have been planned and established.

g. Changes to baselines are controlled (reviewed, analyzed, recorded, approved, and tracked to closure)

h. Software Work Products are released only from the baseline area.

i. An auditor who is independent of the project shall conduct the Configuration Audits using CM Audit Check List

9.1.4 Access to Configuration area

a. Admin access: Administrative access to the configuration area will be granted to the Tech Lead and IT team, while access will be restricted for all other individuals.

b. Read access: All team members who are part of the project will be granted read access.

c. Write access: Write access will be granted to the Project Manager, Team Lead, and Developers, while it will be restricted for Testers.

d. Delete access: Delete access will be restricted for all members except the Project Manager and Team Lead

e. All Confidential documents will be stored in a separate folder and will only be accessible to the Project Manager or person designated by Project manager.

f. Customer Configuration Area: The CM or PM will have Write Access and Team members will have read access. Note: Deviation shall be taken for the projects which use Client's repository for storing all project related artifacts however a SVN repository to be created to store the basic project documents

9.1.5 Version Control

a. Version control can be used for Configuration reporting.

b. Version Control shall be implemented to identify, segregate the CIs as mentioned in the Project Plan.

c. Version Control shall be implemented as defined in the Project Plan.

d. Version Control shall enable to distinguish between the latest and older versions of the CIs and prevent unintended use of non-conforming CIs.

e. Version control can be used for Configuration reporting.

10. Information Security requirements

a. Confidentiality of data is maintained.

b. Changes to the organization, business processes, information processing facilities and systems that affect information security shall be controlled.

c. Changes to systems within the development lifecycle shall be controlled using formal change control procedures.

d. When operating platforms are changed, business critical applications shall be reviewed and tested to ensure there is no adverse impact on organizational operations or security.

e. Modifications to software packages shall be discouraged, limited to necessary changes and all changes shall be strictly controlled.

11. References

a. ISO 9001 Clauses 6.3

b. ISO 27001 Clauses 8.1, Controls A.14

c. ISMS Control Policies

d. SVN Folder Structure UT/PS/CMP/GDL/01

e. Project Planning (PPP)-UTPMPPPSOP

f. Management Systems (MSP)-UTIMMAISOP

12. Annexures

a. NIL

13. Forms

a. Baseline Register UT/PS/CMP/TMP/02

b. CI & CSI List UT/PS/CMP/TMP/04

c. CM Audit Checklist UT/PS/CMP/TMP/05